

Comparative Evaluation of Machine Learning Models for Privacy-Preserving Record Linkage in Real-World Healthcare Data

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Introduction

- Clinical data are often fragmented across healthcare systems, limiting a complete view of patients
- Traditional record linkage relies on identifiable protected health information (PHI), raising privacy and security concerns
- Privacy-Preserving Record Linkage (PPRL) enables secure matching using transformed or encrypted identifiers instead of raw PHI
- The comparative performance of machine learning approaches for PPRL in real-world data remains underexplored
- This study evaluates multiple machine learning models for PPRL and validates performance differences using statistical testing

Practicum Scope

- Designed and evaluated a privacy-preserving framework for patient record linkage using real-world healthcare data
- Built a secure pipeline for record matching without direct exposure of sensitive identifiers
- Analyzed how data representation and preprocessing impact linkage performance in real-world settings
- Examined trade-offs between matching accuracy and error risks under different evaluation conditions

PPRL enables:

- Secure linkage without revealing personally identifiable information
- Compliance with data privacy regulations
- Integration of fragmented healthcare data across systems
- Reduction of duplicate records and improved patient identity resolution
- Support for multi-institutional research and analytics
- Handling of data inconsistencies (missing values, identifier variations)
- Reduced reliance on manual matching through scalable ML approaches
- Improved longitudinal patient tracking and continuity of care
- Enhanced population-level analysis using linked datasets

Preceptor Details

Organization Name: Regenstrief Institute.

Preceptor: Dr. Shaun J. Grannis. **Worksite:** Indianapolis, IN

Mission: Improve healthcare delivery through data-driven research, interoperability, and health information exchange.

Work Environment: Research-focused environment specializing in patient matching and healthcare data integration using privacy-preserving approaches. The practicum involved working with real-world healthcare data in a secure environment, where identifiers were transformed into privacy-preserving representations.

Acknowledgement: I would like to express my sincere gratitude to Dr. Shaun J. Grannis at the Regenstrief Institute for his guidance, support, and valuable feedback throughout this practicum.

Learning Objectives

- Understand privacy-preserving record linkage (PPRL) concepts
- Apply machine learning models to healthcare data
- Handle class imbalance in classification tasks
- Evaluate models using precision, recall, and F1 score
- Perform statistical validation (McNemar, paired t-test, Wilcoxon)
- Interpret and communicate analytical results

Timeline

- **February 2026**
 - Data understanding, preprocessing, and handling masked/missing values
 - Feature engineering (binary agreement features)
- **March 2026**
 - Exploratory analysis and class imbalance handling
 - Model development (SVM, KNN, Logistic Regression, SNN, MLP, XGBoost)
 - Model evaluation (accuracy, precision, recall, F1 score), confusion matrix, and threshold tuning
- **April 2026**
 - Statistical testing (McNemar, paired t-test, Wilcoxon signed-rank test)
 - Result interpretation and validation

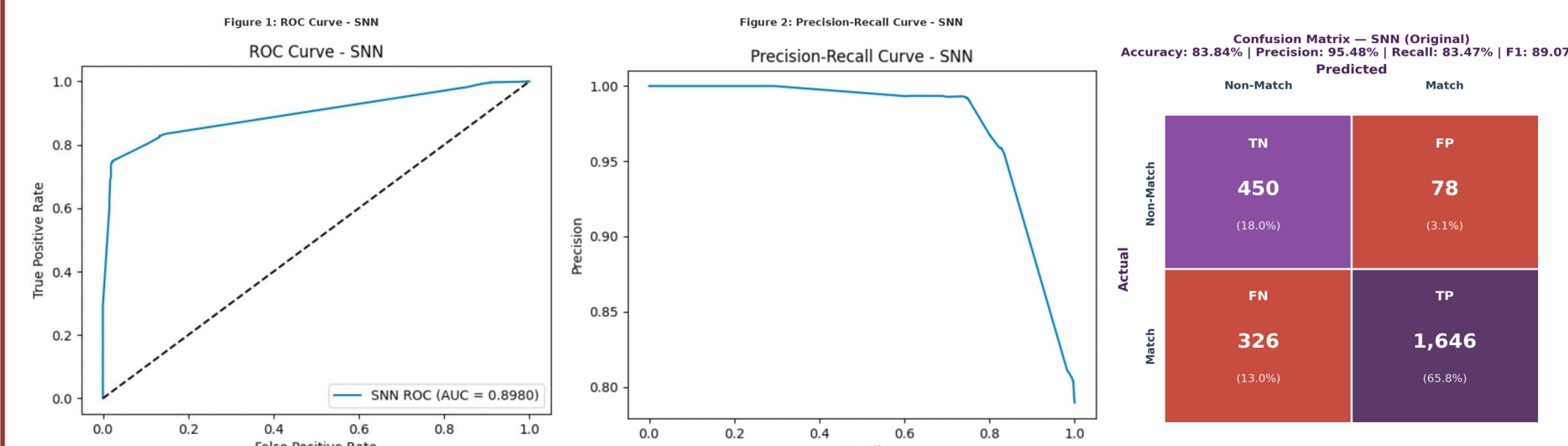
Practicum Duties

- **Task:** Evaluate machine learning models for privacy-preserving record linkage using real-world healthcare data.
- **Dataset:** 10,000 record pairs derived from real-world healthcare data. Each pair was represented using binary agreement features (match = 1, no match = 0). Features were generated from masked identifiers to preserve patient privacy. Class imbalance was present (higher number of matches than non-matches).
- **Methodology:**
 - Preprocessed dataset by handling masked and missing values
 - Generated binary agreement features for record pair comparison
 - Explored dataset characteristics, including class imbalance and feature distributions
 - Split dataset into 75% training and 25% testing sets for model development & evaluation
 - Applied dimensionality reduction (PCA) to improve efficiency
 - Trained & compared multiple models: XGBoost, SVM, KNN, Logistic Regression, SNN, MLP
 - Applied class imbalance handling techniques (upsampling, downsampling, SMOTE)
 - Tuned model parameters and adjusted decision thresholds
 - Evaluated models using accuracy, precision, recall, F1 score, and confusion matrix
 - Compared model performance across different sampling strategies
 - Performed statistical testing (McNemar, paired t-test, Wilcoxon signed-rank test)
 - Interpreted results and analyzed precision–recall trade-offs
- **Tools:** Python, Pandas, NumPy, Scikit-learn (SVM, KNN, Logistic Regression, metrics, PCA), XGBoost, SciPy (McNemar, paired t-test, Wilcoxon), Imbalanced-learn (SMOTE, resampling), Matplotlib (visualization), Custom Neural Network implementation.

Table 1: PPRL Model Performance Results (Best Configuration per Model)

Model	Sampling	Accuracy	Precision	Recall	F1	TN	FP	FN	TP
SNN	Original	0.8384	0.9548	0.8347	0.8907	450	78	326	1646
MLP	Original	0.8380	0.9542	0.8347	0.8905	449	79	326	1646
SVM	Original	0.8380	0.9542	0.8347	0.8905	449	79	326	1646
Logistic Regression	Downsample	0.8236	0.9659	0.8048	0.8780	472	56	385	1587
KNN	Original	0.8056	0.8111	0.9823	0.8885	77	451	35	1937
XGBoost	Upsample	0.8224	0.9659	0.8032	0.8771	472	56	388	1584

Best overall (SNN Original)



Practicum Outcomes – Professional

- Demonstrated that multiple machine learning models achieve similar performance in PPRL tasks
- Found that simpler models (e.g., Logistic Regression, SVM) perform comparably to more complex models
- Observed that XGBoost showed comparatively lower performance on binary agreement features. Identified the impact of class imbalance on recall and overall model behavior.
- Learned that threshold tuning significantly influences precision–recall trade-offs. Validated that performance differences between models are not always statistically significant
- Established that privacy-preserving record linkage can achieve effective matching performance without direct access to sensitive patient identifiers

Figure 3: F1 Score Comparison Across Models

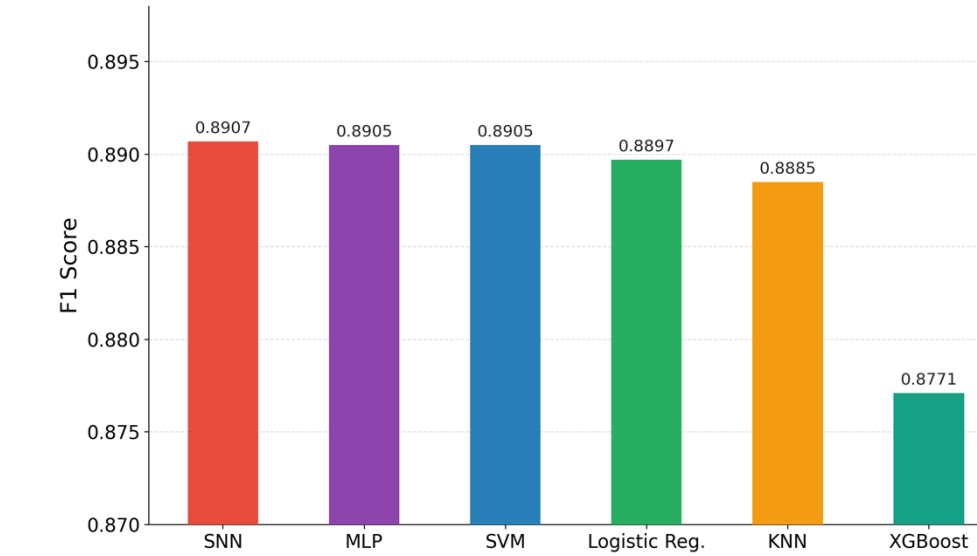
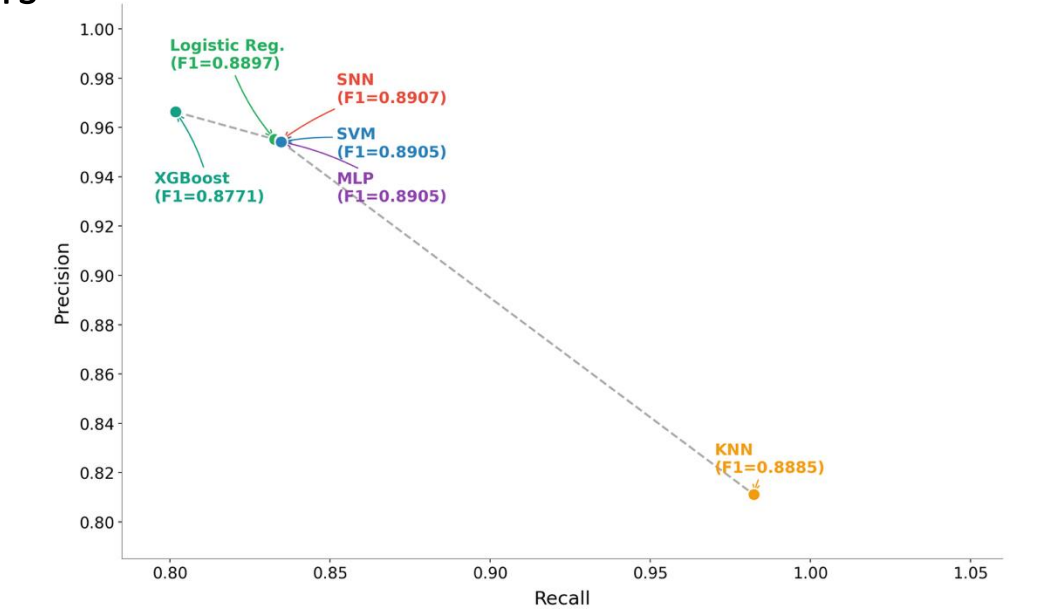


Figure 4: Precision-Recall Trade-off Across Models



Practicum Outcomes – Learning Objectives

- SNN achieved the highest performance (F1 = 0.8907), demonstrating the best precision–recall balance. SVM, Logistic Regression, and MLP showed comparable performance (F1 ≈ 0.889–0.890).
- XGBoost showed comparatively lower performance across sampling strategies, likely due to the limited non-linear complexity of binary agreement features
- Statistical tests confirmed that the difference between SNN and XGBoost was significant, while top-performing models showed no statistically significant differences
- Sampling strategies improved XGBoost performance but did not change the overall model ranking. KNN demonstrated high recall but inconsistent precision, indicating variability in performance
- Precision–recall trade-offs highlight the importance of minimizing false matches in healthcare applications.

Table 2: Statistical Significance Tests: McNemar | Paired t-test | Wilcoxon Signed-Rank

Comparison	McNemar p-value	t-test p-value	Wilcoxon p-value	Significant?	Better Model
SNN vs XGBoost	0.00019	0.000165	0.000168	✓ Yes	SNN
SNN vs SVM	1.00000	0.317407	1.000000	No	Same
SNN vs MLP	1.00000	0.317407	1.000000	No	Same
SNN vs Logistic Regression	0.45312	0.256921	0.375000	No	Same
SNN vs KNN	0.00165	0.001450	0.001462	✓ Yes	SNN
XGBoost vs SVM	0.00022	0.000189	0.000192	✓ Yes	SVM
XGBoost vs Logistic Regression	0.00029	0.000252	0.000256	✓ Yes	Logistic Regression

Significant (p < 0.05) | Not significant

Conclusion

- Machine learning models effectively perform privacy-preserving record linkage using transformed healthcare data, with SNN and other models showing comparable performance.
- Performance differences were influenced more by data representation and class imbalance than model complexity, suggesting simpler models (e.g., Logistic Regression, SVM) may be preferred for interpretability.
- Precision–recall trade-offs emphasize the need to minimize false matches, highlighting the importance of careful model selection, class imbalance handling, and rigorous evaluation, with future work focused on improving robustness and scalability.



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